Study on effectiveness in real-world with dapagliflozin and sitagliptin fixed dose combination in patients with type 2 diabetes mellitus in India: Retrospective study from electronic medical records

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Introduction

- Type 2 diabetes mellitus is a chronic, progressive metabolic disorder with high prevalence in India (101 million in 2021)[1,2] with more than 75% patients having uncontrolled diabetes (HbA1c ≥ 7%) [3]
- Since glycemic control deteriorates over time, treatment intensification with multiple OADs is often required in patients inadequately controlled with monotherapy [4]
- FDCs with complementary mechanisms of actions can provide sustained glycemic control; help in reducing pill burden and dosing frequency, lowers cost of therapy and improves compliance [5,6]
- Combination therapy of Dapagliflozin and Sitagliptin is a novel treatment option that exhibits synergistic effect [7,8]
- SGLT2i increase glucagon concentration and hepatic glucose production while DPP4i stimulate postprandial insulin release and inhibit glucagon release. Moreover, Dapagliflozin has shown to have cardiorenal benefits while Sitagliptin helps in beta cell preservation.[8]

Benefits of Dapagliflozin and Sitagliptin FDC in Indian patients[9,10,11,12]

DPP4i
- Inhibits DPP-4 enzyme and thereby prevents degradation of GLP
- Demonstrated better response in Asians
- Beta cell preservation and well-tolerated

SGLT2i
- Glycemic control
- Weight reduction
- Improvement in beta cell function
- Cardiorenal benefits

Aim: A real-world, retrospective, observational, EMR-based study to asses effectiveness and usage pattern of Dapagliflozin and Sitagliptin FDC in Indian patients with T2DM

Method

- In this study aggregated and anonymized EMR data from the patients meeting the eligibility criteria was retrieved and analyzed. Study was approved by Independent Ethics Committee and registered on CTRI. (CTRI number-CTRI/2023/10/058365).

Patients prescribed with Dapagliflozin and Sitagliptin FDC as captured in EMR (2022–2023)

EMR database → Patients meeting eligibility criteria → Visit 1 (Baseline visit) → Visit 2 (3 ± 1 months from baseline) → Visit 3 (6 ± 1 months from baseline)

Parameters – Demographic characteristics, symptoms, diagnosis, HbA1c, glucose levels, comorbidities, medication details

Inclusion criteria	Exclusion criteria
Patient with • Age ≥ 18 years • Diagnosed with T2DM • HbA1c ≥ 7% • Prescribed Dapagliflozin and Sitagliptin FDC in any visit other than baseline visit on the EMR platform • EMR data captured for more than one visit	• T1DM • Patients on insulin or any other injectable antidiabetic medication • Patients with incomplete medical records with respect to the required study parameters

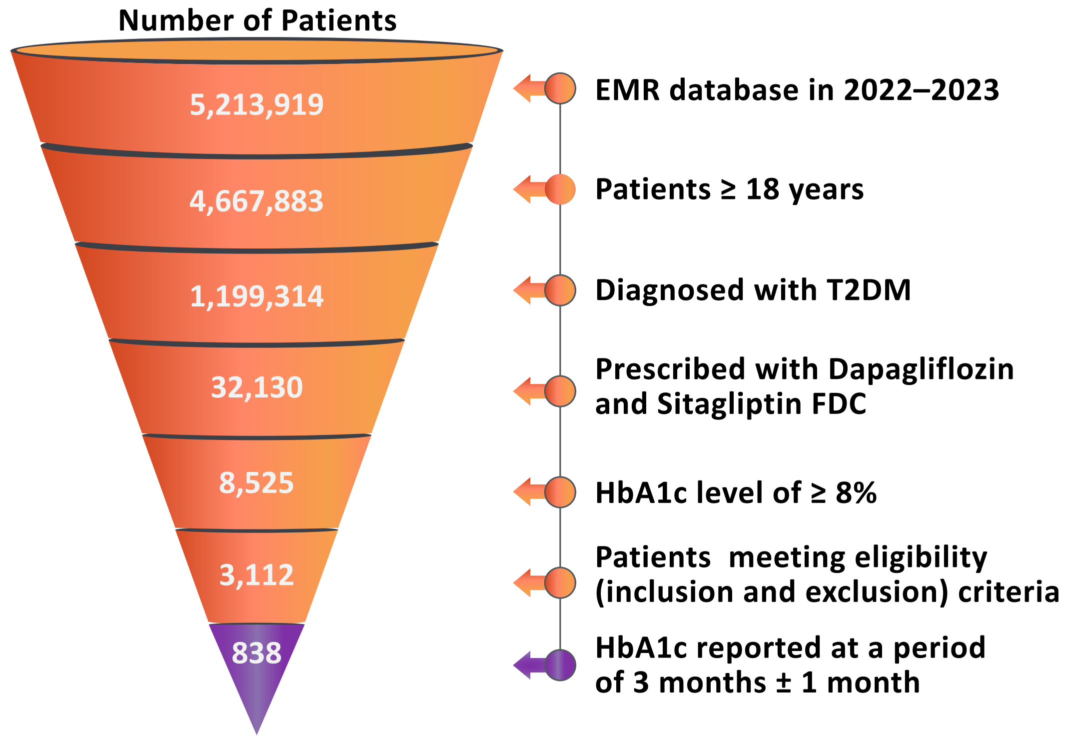
Primary endpoint

Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥ 8% at baseline

Key secondary endpoints

- Mean change in HbA1c, FBG, and PPG from baseline to 3 months and 6 months in patients with HbA1c ≥ 8% at baseline
- Mean change in HbA1c from baseline to 3 months and 3 months and 6 months in patients with HbA1c ≥ 7% at baseline
- Mean change in HbA1c from baseline to 3 months in patients with HbA1c ≥ 8% at baseline, upon stratification of patients by age, gender, weight/BMI, baseline OAD therapy, dose of Dapagliflozin and Sitagliptin FDC and comorbidities (cardiovascular, renal, and others)

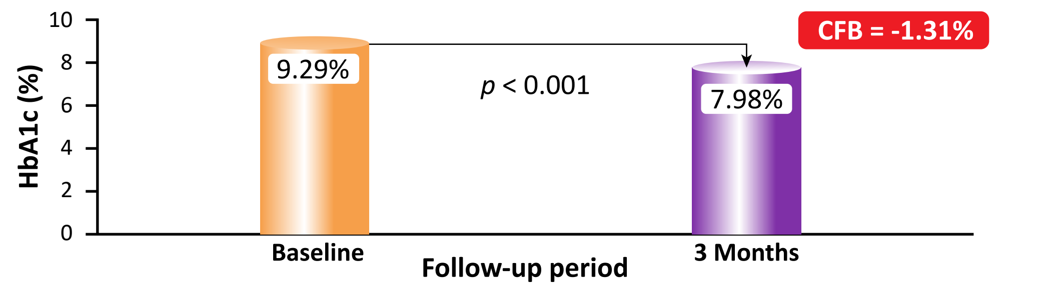
Results



Primary endpoint

A significant reduction in mean HbA1c was observed from baseline to 3 months (Figure 1)

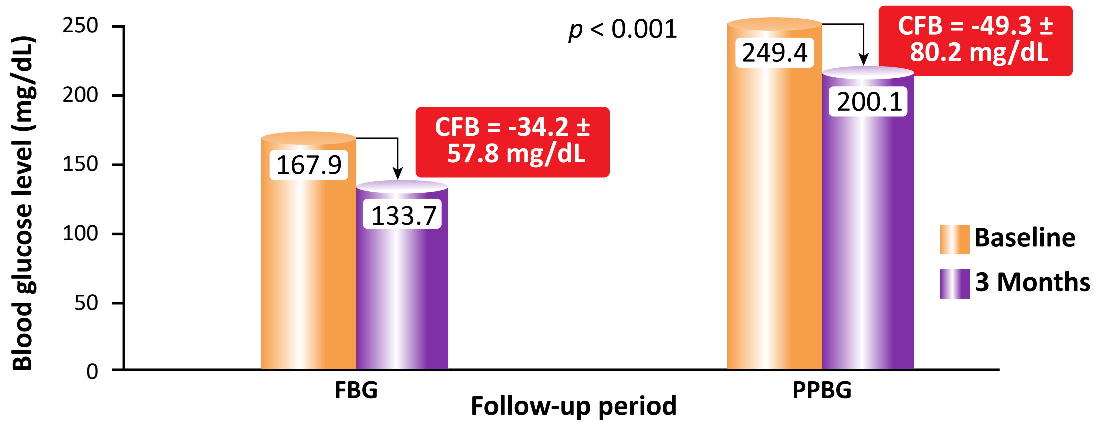
Figure 1: Mean change in HbA1c from baseline to 3 months



Key secondary endpoints

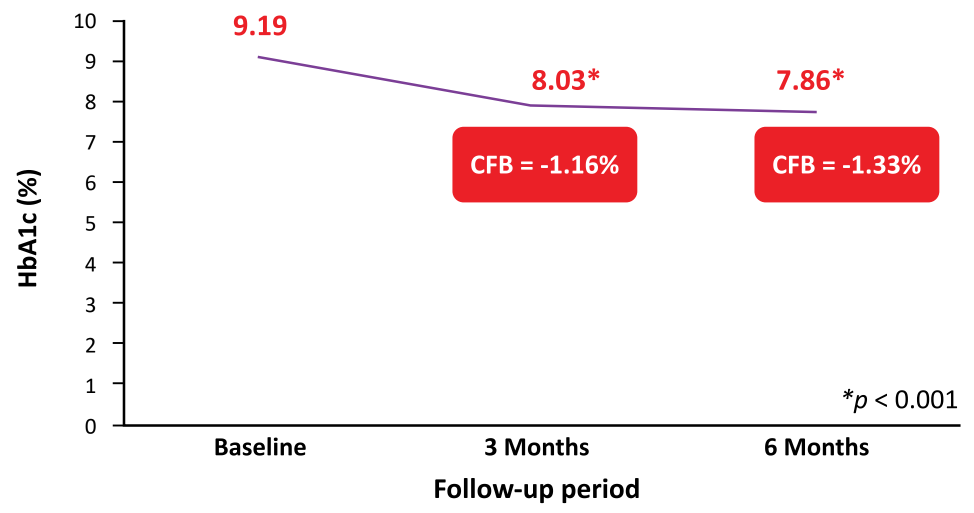
A significant reduction in FBG (N = 710) and PPBG (N = 622) was observed from baseline to 3 months (Figure 2)

Figure 2: Mean change in FBG and PPBG from baseline to 3 months



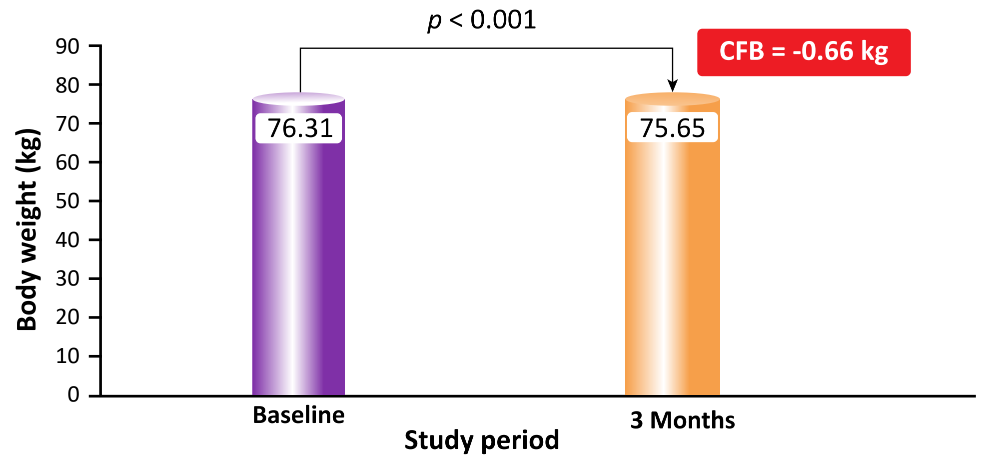
Significant reduction in HbA1C was observed from baseline to 3 months and 6 months (N = 146, Figure 3)

Figure 3: Mean change in HbA1c from baseline to 3 and 6 months



- In patients with baseline HbA1c ≥ 7%, a significant reduction in HbA1c was observed at 3 months (N = 1,418) and at 3 and 6 months (N =261)
- A significant reduction in HbA1c was observed irrespective of age, gender, BMI, and presence of CV comorbidities
- In patients with baseline HbA1c ≥ 8% and BMI > 25 kg/m^2 (N = 649), there was a significant reduction in body weight from baseline to 3 months (Figure 4)

Figure 4: Mean change in body weight from baseline to 3 months



Study limitations

- This a single arm analysis of EMR data captured on digital platform
- The database on which this study was conducted, does not capture safety data

Conclusion

- In patients with T2DM and HbA1c ≥ 8% at baseline, FDC of Dapagliflozin and Sitagliptin showed significant improvement in HbA1c levels
- Dapagliflozin and Sitagliptin FDC effectively reduced HbA1c, FBG, and PPBG in patients with T2DM in 3 months, indicating effective overall glycemic control
- The FDC was also effective in patients irrespective of patients' age, gender, BMI, and presence of CV comorbidities. In addition, it reduced body weight
- Thus, FDC Dapagliflozin + Sitagliptin was shown to be effective treatment option in patients with T2DM in real-world setting

Acknowledgements

Authors would like to thank HealthPlix for providing data source.

Abbreviations:

BMI, body mass index; CTRI, clinical trials registry-India; CV, cardiovascular; DPP4i, dipeptidyl peptidase 4 inhibitor; EMR, electronic medical record; FDC, fixed dose combination; FBG, fasting blood glucose; HbA1c, glycated hemoglobin; OAD, oral antidiabetic drug; PPBG, postprandial blood glucose; SGLT2i, sodium-glucose cotransporter-2 inhibitor; T1DM, type 1 diabetes; T2DM, type 2 diabetes

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